Memo No.
Date / /

1. (a).

i. When $k_i^T q$ ~~takes~~ $\geq k_j^T q$ For any other $j \neq i$.

ii. ~~Fo~~ In condition in i, $c = v_i$

(b). Set the query $q = \beta(k_a + k_b)$ where β is a big constant.

$$k_a^T q = k_a^T \cdot \beta(k_a + k_b) = \beta(1+0) = \beta$$

$k_b^T q = \beta$ similarly.

For other terms that $i \neq a$ or b.

$$\alpha_a = \frac{e^\beta}{e^\beta + e^\beta + (j-2)e^0}$$

When β is big enough, $\alpha_a \approx \frac{1}{2}$.

Proved.

(c). q

i. $q = \beta(\mu_a + \mu_b)$. , β ~~nee~~ should be big

ii. Bc. $k^T q$ is passed through a exponential function. any disturburce will cause huge eff

Mo | Tu | We | Th | Fr | Sa | Su

Memo No. ______
Date / /

(d) i.

$q_1 = \cancel{\beta k_1}\ \beta \mu_1$; $q_2 = \beta \mu_2$.

β is a big constanct acc. to (c).

ii. c_1 will basically focuses on k_a and c_2 on v_b.

$\frac{1}{2}(c_1 + c_2)$ will approach $\frac{1}{2}(\cancel{k_a + l}\ v_a + v_b)$ with a great variance.

(e) it overcomes the drawback of not being able to handle multiple sources of info.

2. (a). i

$\cancel{Q}$. $H_{perm} = \mathrm{softmax}\left(\frac{PXW_Q W_K^T X^T P^T}{\sqrt{d}}\right) PXW_V$

$= \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) PXW_v.$

$Z_{perm} = \mathrm{ReLu}(H_{perm} W_1 + b_1) W_2 + b_2$

$= PZ.$

ii That means transformer is not able to understand the permutation of tokens, which may be problematic for NLP.

Memo No.
Date / /

(b). i

Yes, the Pos embeddings are added to the vectors, making each vector different.

ii.

No, suppose the Pos embeddings are the same for ~~differen~~ two identical token in differernt position. Transformer will not be able to specify it.

扫描全能王 創建